

Counterparty Credit Risk Engine

Final Presentation: Data, Methods, Trade-offs, and Results

Berkeley MFE Industry Project – Capitolis

September 30, 2026

Agenda

Project Overview

Data: Sources, Uses, and Why Real Data Matters

Engine Architecture

Risk-Factor Models and Calibration

Simulation Methodology

Headline Results

Sensitivities (Greeks)

Stress Testing

Model Validation and Model Risk

Credit and Capital: xVA

Feedback Addressed

Conclusions

The book, and the objective

16 trades, 3 counterparties

- 8 equity total-return swaps (incl. 2 JPY-compo)
- 4 bond forwards, 4 bond total-return swaps
- 37 equities (US + Tokyo) + USDJPY
- Capitolis is strictly the seller on every trade

Objective (from the kickoff brief)

- Build a modular Monte Carlo CCR engine from scratch
- Stochastic models → pricer integration → exposure & Greeks → performance
- Extra credit: xVA; risky bonds + CDS

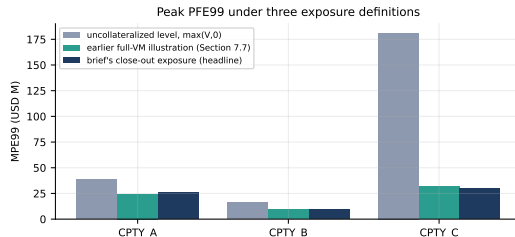


Trade life spans: nearly all exposure disappears within 4 months; only 2 trades run past mid-2027.

Exposure definition, and why two numbers

$$\text{Close-out: } \max(V(t+10\text{bd}) - V(t-1\text{bd}), 0) \quad \text{Level: } \max(V(t), 0)$$

- **Close-out** (the brief's definition, primary): signed prior-day variation margin, no IM/threshold, the 10-day move from an already-margined start, netted per counterparty
- **Level** (secondary): whole mark-to-market, no margin at all – a no-margin upper bound
- Real answer sits between the two; the actual CSA's threshold/MTA/IM were not supplied



- **Every number in this project traces to a disclosed, real source** – no synthetic placeholders for calibration inputs
- **Benefit:** results are reproducible, auditable, and defensible in front of a risk committee; every assumption is a named simplification, not a silent one
- Two tiers: **public** (Yahoo Finance, FRED, Databento) and **licensed** (Bloomberg, used and cross-checked but never redistributed as raw series)

Data source 1: USD discount curve (Databento SOFR futures)

- **Source:** real Databento CME SOFR futures settlement prices (SR3), bootstrapped ourselves
- **Used for:** every USD discount factor; the base for the Hull-White short-rate model
- **Extension:** Bloomberg long-end quotes spliced in beyond futures coverage (out to 50y), since SOFR futures alone only cover the front few years

Benefit of real data: a genuine market-implied curve, not an assumed flat extrapolation. In the correction-by-correction attribution (see Model Validation section), splicing the real Bloomberg long end onto the previously-flat curve moved the portfolio MPE99 from \$52.7M to \$52.1M (−1.1%) – a modest, disclosed effect, not a large one, but a genuinely measured one rather than an assumption

Data source 2: Treasury yields (FRED) and swaption vols (Bloomberg)

Source	Used for	Benefit of using it
FRED Treasury CMT yields (3mo–30y, public, daily since 1981)	Realised long-end yield volatility, used to fit Hull-White σ (0.96%); Kupiec backtest of the rate model	Directly measures the volatility of the yields this book's bond positions actually depend on, not a proxy
Bloomberg ATM normal swaption vol cube	Mean-reversion a (0.0167, $R^2=0.84$), via the decay of vol with tenor	Textbook calibration route for a ; genuinely implied, forward-looking

Data source 3: equities, FX, JPY curve

Source	Used for	Benefit of using it
Yahoo Finance: 37 equity closes + dividends, 3y history	Equity spots, dividend yields, realised volatility, the 39×39 correlation matrix	Public, free, long enough history (2014–today, extended to 2007 for stress) to estimate realised statistics reliably
Yahoo Finance: USD-JPY spot, 3y history	FX spot, realised FX volatility, FX–equity correlation	Same realised-statistics benefit, consistent methodology across all factors
Real JPY OIS curve (public JPY market data)	JPY Hull-White factor's base curve	Lets JPY rates move independently and go negative, matching the real BOJ history

Data source 4: credit spreads, and the licensed Bloomberg data

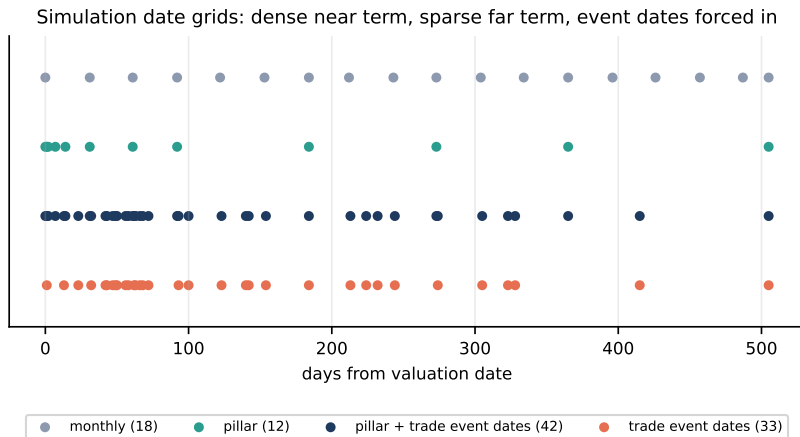
Source	Used for	Benefit of using it
ICE BofA bond-index OAS by rating (FRED, public)	Counterparty and issuer credit spread proxy (CVA, DVA, risky-bond pricing)	No CDS available; this is the Basel MAR50.32(3)-sanctioned fallback – real market spreads by credit quality, not an as- sumed flat number
Bloomberg (licensed): long-end curve, USD/JPY swaption vol cubes, FX vol sur- face	USD/JPY curve splice, mean-reversion fit, model-risk cross-checks	The only source for genuine long-dated mar- ket data beyond what's public; used inter- nally and only derived numbers are ever re- ported (never raw series), per the license

Benefit, summarised: every model parameter is either directly measured or transparently proxied from real data, so every number in this deck can be traced back to a named, dated market source.

The simulation pipeline

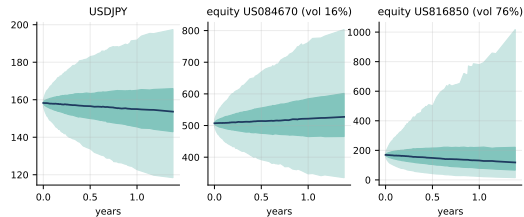
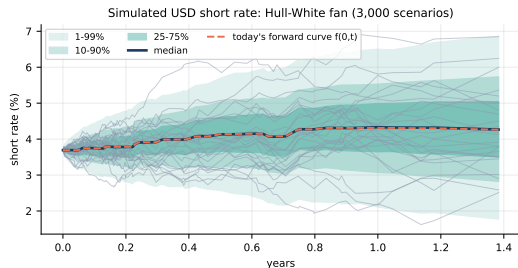
1. **Calibrate:** pull real data → fit every model parameter (previous section)
2. **Draw:** Latin Hypercube random numbers → correlated shocks (Cholesky decomposition of the real 39×39 matrix)
3. **Simulate:** step every risk factor jointly across the full time grid (equities, FX, USD rate, JPY rate)
4. **Reprice:** feed simulated market states into the provided pricers – every trade, every date, every scenario
5. **Aggregate:** net by counterparty → EE / median PFE / PFE99 / MPE, on both exposure conventions
6. **Downstream:** Greeks, stress, CVA/DVA/FVA/KVA, SA-CVA, SA-CCR – all reusing the same simulated paths

Simulation grid design: dense near term, sparse far term



The pillar grid (12 dates) plus forced trade-event dates (42 total) is more compact than a plain monthly grid (18 dates) and hits every real cash-flow date exactly – fewer nodes to reprice, no loss of precision at the dates that matter.

What the simulated paths actually look like

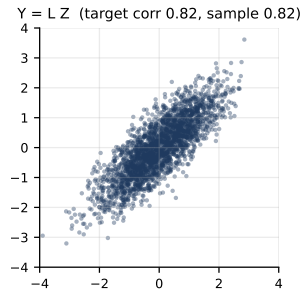
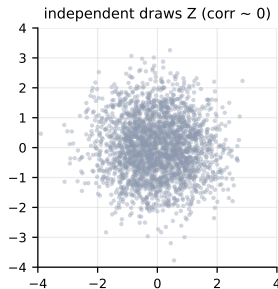


Left: simulated USD short rate (median tracks today's forward curve, since the model is calibrated to reproduce it exactly).

Right: simulated USDJPY and two equities with 1–99% and 25–75% percentile bands.

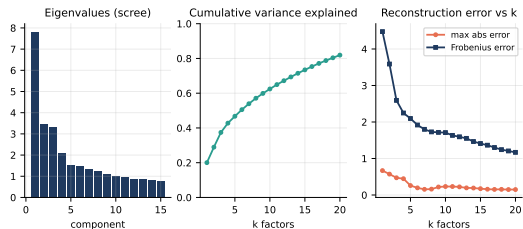
Equity and FX: correlated geometric Brownian motion

- Standard lognormal diffusion, risk-neutral drift ($r - q$), correlated via the real 39×39 matrix
- JPY-quoted names carry a quanto correction; USDJPY drift enforces the martingale condition on the JPY money-market account valued in USD



Cholesky demonstration on the most correlated real pair (corr 0.82): independent draws (left) become correlated (right) after multiplying by L .

Correlation: full Cholesky matrix vs. PCA factor model



		Full	PCA k=5	PCA k=10
Step	time	1.91s	2.08s	2.29s
(5,000 scen.)				
Variance	ex-	100%	47%	62%
plained				
Netting-set	vol	–	B +1.7%	B +1.5%
vs. full				

Selected full matrix: PCA is *slower*, not faster, at $n=39$ (draws

k systematic + n idiosyncratic random numbers, no dimensionality win). **Loss of using PCA instead:** no speed gain, and up to 1.7% extra sampling error – a pure loss here. Kept only as an explainability tool: factor 1 (20% of variance) is a broad market mode, factor 2 (8.9%) is Japan vs. US.

USD rates: calibrating mean reversion (a) – two real routes compared

Route	Data	a	R^2	Verdict	
Swaption cube (preferred)	Bloomberg ATM normal vol, 1M expiry, 1–15Y tenors	0.0167	0.845	Selected used	Both
SOFR-futures vol decay	~2y Databento history, 8 contracts	0.0458	0.820	cross-check only	

real-data routes give sensible, positive values with similar fit quality; the swaption route is the textbook standard (implied, not realised) and was preferred. The two disagree because they use different instruments (implied option vol vs. realised futures moves) and different tenor ranges – a real, disclosed source of model uncertainty, quantified directly in the model-risk study ($a \times 3$ moves the headline number by -5%).

USD rates: calibrating volatility (σ) – realised, and which window

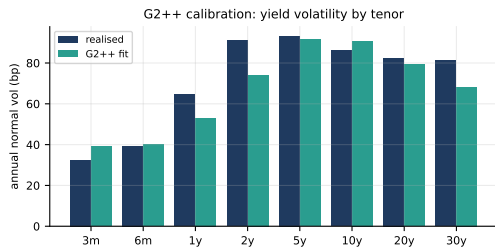
- **Original source:** realised volatility of the overnight SOFR fixing itself, 0.63%
- **Problem:** overnight fixings move in discrete steps on Fed days – understates how far the *long-dated* yields this book depends on actually move (realised 10-day change of the 20y yield: $\sim 15\text{bp}$ std dev, vs. $\sim 11\text{bp}$ implied by the overnight sigma)
- **Selected Changed to:** realised volatility of 2y–30y Treasury yield changes directly, 0.96%

Validated by Kupiec backtest (99% VaR, rate moves, 161 ten-day windows, 3 tenors):

Tenor	Overnight-SOFR σ : up/down exc.	Long-end σ : up/down exc.	Expected	Both pass?	
5y	1 / 1	4 / 3	1.6	yes ($p > 0.05$)	Both sigma
10y	1 / 0	3 / 2	1.6	yes	
20y	1 / 1	3 / 1	1.6	yes	

choices pass Kupiec at this sample size; the switch to long-end realised vol is motivated by the direct realised-move comparison above, which is the more targeted diagnostic for this book's long-dated exposure.

USD rates: one-factor vs. two-factor Hull-White

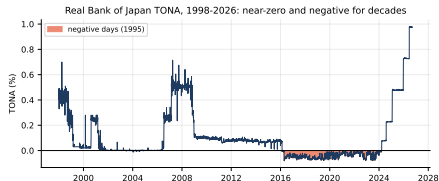


	1-factor	2-factor (G2++)
Fit to realised yield covariance	–	16.1% relative error
Parameters	$a=0.0167$	$\sigma=1.21\%$, $a=0.039$, $\eta=1.44\%$, $b=1.5$ (at bound), $\rho=-0.95$
Effect on portfolio MPE99	baseline \$51.5M	-3.0% (A -1% , B -3% , C -0%)

Benefit of 2-factor: lets the curve twist, not just shift – matters if

exposure spans many maturities. **Cost:** a parameter pinned at its bound, 16% fit error, and a 3% move that doesn't justify the complexity here. **Selected:** one-factor. The book's rate risk is one large single-maturity bond forward – a level bet, not a curve-shape bet.

JPY rates: a dedicated factor vs. a constant differential



Alternative rejected: JPY names/USDJPY drift off one constant USD–JPY rate differential – cheap, but cannot represent JPY rates moving independently, or negative.

Selected: a second, real, simulated Hull-White factor from the JPY OIS curve. TONA has been negative or near-zero for most of 25 years – a floored model cannot represent this. **Mean-reversion calibration – all 3**

real routes tried, all invalid:

Route	Fitted a	
JPY OIS history	−0.0261	All
Swaption cube	−0.0374	
JGB yield history	−0.0203	

negative: JPY vol genuinely *rises* with tenor. Used the lowest admissible value ($a=0.001$, Ho-Lee limit) rather than an invalid fit. **Result:** moved

portfolio MPE99 attribution \$52.1M → \$51.5M – real, disclosed, not noise.

Sampling scheme: five methods, exact benchmark numbers

Controlled test (European call, known Black-Scholes truth = 41.077):

Method	RMSE	vs. pseudo-random	ms/trial
Pseudo-random	1.456	1.0x (baseline)	0.30
Antithetic	1.079	1.3x better	0.31
Moment-matched	0.232	6.3x better	0.55
Sobol	0.051	28.8x better	1.65
Latin Hypercube	0.029	50.6x better	0.98

Real 663-dimensional engine check (std. dev. of PFE /

median, across repeats): no method wins both – Sobol has the lowest PFE std (2.72M vs. 4.29M pseudo-random), but moment-matched has the lowest median std (0.40M vs. 0.76M pseudo-random); Latin Hypercube is competitive on both (3.73M / 0.48M) without being worst on either. **Selected** Latin Hypercube: best on the controlled test, competitive (never worst) on both real-engine statistics, negligible extra cost.

Scenario count: exact convergence results

Close-out MPE99, relative standard error by bootstrap (peak date 2026-09-20):

Netting set	N=250	N=500	N=1,000	N=2,000	N=3,000
CPTY_A	6.3%	5.1%	3.2%	2.6%	2.5%
CPTY_B	5.1%	4.7%	3.8%	3.5%	3.5%
CPTY_C	5.9%	4.8%	3.8%	3.1%	3.1%
Portfolio	7.1%	4.9%	3.3%	2.2%	2.2%

Portfolio relative SE at $N=2,000$ is 2.2%,

extrapolating ($1/\sqrt{N}$) to $\sim 1.4\%$ at $N=5,000$ and $\sim 1.0\%$ at $N=10,000$. **Independent 30,000-scenario pool study (1y-node**

N		1,000	5,000	10,000	20,000	30,000 (pool)
PFE99, level exposure):	Rel. SE	0.66%	0.29%	0.21%	0.09%	0 (reference)
	Est. time (8 cores)	119s	479s	929s	1,829s	2,729s

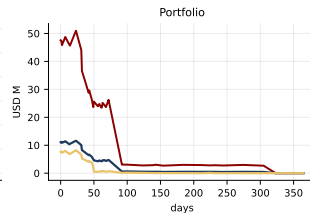
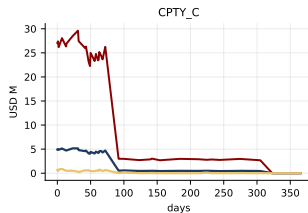
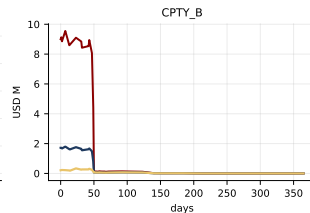
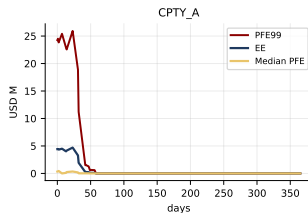
Selected: $N=5,000$

for standard reporting (good precision, ~ 8 min); $N=10,000$ for sign-off (~ 15 min); beyond $\sim 20,000$, marginal SE improvement per extra 5,000-scenario batch falls below 30% while cost keeps scaling linearly – diminishing returns, and remaining error is already smaller than model-risk sensitivities (later section).

Headline results (5,000 scenarios, close-out exposure)

	MPE99	Peak EE
CPTY_A	\$25.9M	\$4.7M
CPTY_B	\$9.6M	\$1.8M
CPTY_C	\$29.6M	\$5.2M
Portfolio	\$51.0M	\$11.6M

- CVA \$11.7k close-out / \$247k level
- SA-CVA capital \$0.10M / \$2.23M
- SA-CCR EAD \$260M
- KVA (10% CoC) \$22.4k / \$470k
- Level exposure MPE99: \$201.9M



What drives each counterparty

	Equity $\Delta/+1\%$	FX $\Delta/+1\%$	DV01/ $+1\text{bp}$
CPTY_A	-\$2.25M	\$0	\$31k
CPTY_B	-\$0.96M	+\$0.42M	-\$4k
CPTY_C	-\$1.24M	\$0	\$594k

Rank correlation of exposures: A–B 0.34, A–C 0.33, B–C 0.12

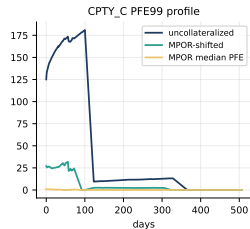
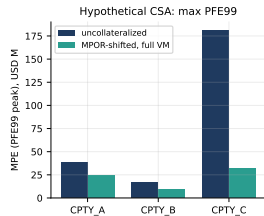
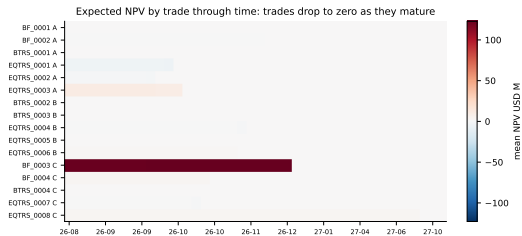
- A is most equity-driven; C is the rate netting set (one \$500M short forward on a 2049 Treasury) – but C also carries meaningful equity delta
- B is the only netting set with material FX delta (holds the 2 JPY-compo swaps)
- Yield-equity correlation ≈ 0.00 in our data – no reliable offset

Why the portfolio is close to additive across counterparties



- Nothing nets *across* counterparties by construction – portfolio MPE99 is a sum of three separate netting sets
- All three counterparties reach their own peak exposure in the same few weeks (before the largest equity baskets mature) – no timing diversification
- Positive rank correlation (0.12–0.34) between netting sets, since every book holds pay-equity swaps
- Result: Portfolio MPE99 (\$51.0M) $\approx$ A+C (\$54.4M), 78% of the simple sum of all three (\$65.1M)

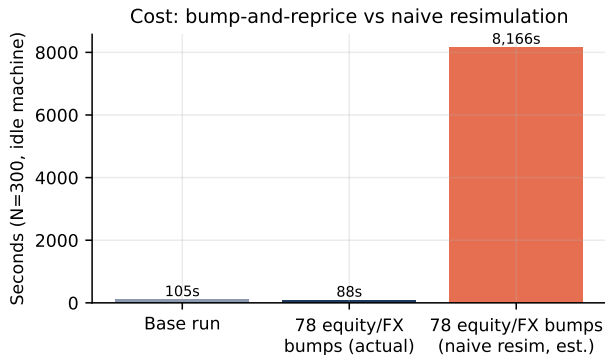
Additional exposure diagnostics



Left: expected NPV by trade – BF_0003 dominates and disappears at its Dec 2026 settlement. Right: peak PFE99 under a hypothetical full-VM CSA, for comparison against the brief's own definition.

Method: common random numbers, and what they buy

- Bumping a risk factor and repricing with **independent** random draws: estimator std. dev. \$245k – too noisy to use
- Reusing the **identical** draws in the base and bumped run (common random numbers, CRN): std. dev. \$1k
- **282x noise reduction** – this, not the sampling scheme, is what actually controls Greeks precision



Equity/FX spot bumps rescale simulated GBM

paths exactly (no resimulation): 78 bumps at N=300 cost 88s actual vs. ~8,166s estimated naive resimulation – **93x cheaper**.

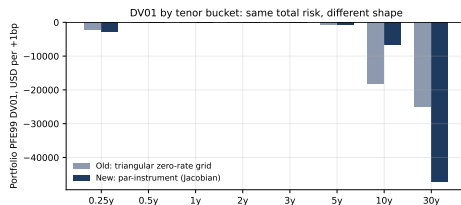
Method comparison: bump-and-reprice vs. pathwise vs. adjoint

Method	Cost	Verdict
Bump, independent draws	baseline, \$245k noise	<i>rejected</i> : noise swamps the signal
Bump, CRN (used for all reported Greeks)	~260s (equity EE, N=300)	<i>Selected</i> : works on black-box pricers, works for quantiles
Pathwise (closed form, equity/FX only)	0.2–0.4s	<i>Selected</i> as fast cross-check: ~1000x faster, agrees to median 0.02% / worst 0.4% vs. bump-and-reprice EE
Adjoint differentiation	would price all Greeks for ~3–5 valuations	<i>rejected</i> : needs pricers rewritten in differentiable form – out of scope, listed as future work

Rate/vol Greeks (13 needed: parallel up/down, 8 buckets, 3 vega) still require a genuine resimulation at ~109s each (N=300, idle machine) – unless the exact analytic method below applies.

Rate DV01: zero-grid bump vs. par-instrument Jacobian – full breakdown

Tenor bucket	Old (zero-grid), USD	New (par-instrument), USD
0.25y	-2,182	-2,892
0.5y	-46	-1
1y	-44	-14
2y	-131	-205
3y	-96	-105
5y	-770	-713
10y	-18,324 (39%)	-6,592 (11%)
30y	-24,988 (54%)	-47,191 (82%)
Sum of buckets	-46,580	-57,713
Independent parallel bump	-46,605	-57,705



Rate DV01: why the new method, and what changed

Old: bump the *fitted* zero curve at 8 hand-picked tenors, triangular interpolation – disconnected from how the curve was actually built.

Selected New: bump the curve's own ~ 45 native construction pillars (real SOFR-futures-implied points + every Bloomberg-spliced long-end point), grouped into the same 8 report buckets by nearest pillar – the genuine par-instrument (Jacobian) sensitivity a rates desk would actually compute. **Material benefit:** the old grid was smearing the book's real 20–50y

rate risk (where BF_0003, the dominant \$500M bond forward maturing 2049, actually sits) partly into the 10y bucket. The new method correctly shows **82% of portfolio DV01 in the 30y bucket**, not 54%. Both methods sum to the same total (internal consistency check, within 0.1%) – the total risk was always right, only the attribution across tenor was wrong.

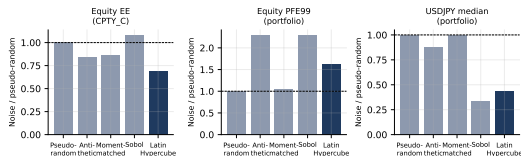
An exact rate Greek, reusing the same simulated draws

In one-factor Hull-White, $r(t) = x(t) + \alpha(t)$: $x(t)$ is the simulated random factor (depends only on random draws, a, σ); $\alpha(t)$ is deterministic (depends only on the curve). A curve bump that leaves a, σ fixed – every DV01 bump here – leaves $x(t)$ **identical** to the base run.

- The whole bump effect is one closed-form number per date, added to the base run's own paths
- **No resimulation. No new random draws.**
- Validated against a true independent resimulation: agreement to 10^{-15} relative precision – not an approximation, the same calculation

Separately: does Latin Hypercube itself reduce Greeks noise? Tested directly – mixed, 15–85% less noise on EE deltas, no

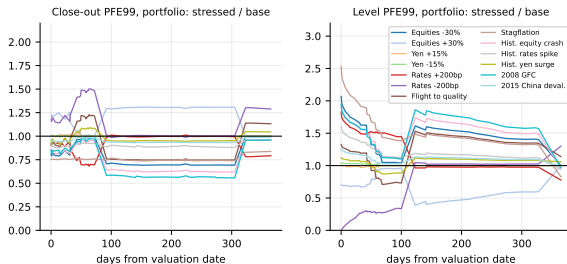
Sampling scheme vs Greeks noise: no consistent winner



consistent benefit on PFE99 deltas:

Design: 13 fixed scenarios, same random draws as base

Family	Scenarios	Shock
Hypothetical (8)	EQ_DOWN/UP_30, JPY_STRONG/WEAK_15, RATES_UP/DOWN_200, FLIGHT_TO_QUALITY, STAGFLATION	Round-number shocks to equities, USDJPY, USD curve
Historical (5)	HIST_EQUITY_CRASH, HIST_RATES_SPIKE, HIST_YEN_SURGE, HIST_GFC_2008, HIST_CHINA_DEVAL_2015	Actual moves over a real 10-day window, chosen by worst-window rule

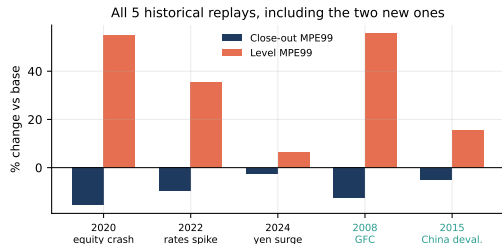


Results: how far each scenario moves the headline number

Scenario	Close-out MPE99	Level MPE99	
EQ_DOWN_30	−15.3%	+62.3%	
RATES_DOWN_200	+22.0%	−69.0%	
STAGFLATION	−24.1%	+89.1% (largest level move)	Close-out exposure is far less
FLIGHT_TO_QUALITY	−2.0%	+19.6%	
HIST_EQUITY_CRASH (2020)	−15.4%	+54.8%	
HIST_GFC_2008	−12.7%	+55.6%	
HIST_CHINA_DEVAL_2015	−4.9%	+15.6%	

sensitive to instantaneous level shocks than level exposure (margin is why). Equity-driven trades react only to equity/FX shocks; bond trades react only to rate shocks, confirming trade structure, not coincidence.

Extending the history: 2008 and 2015, and why it matters

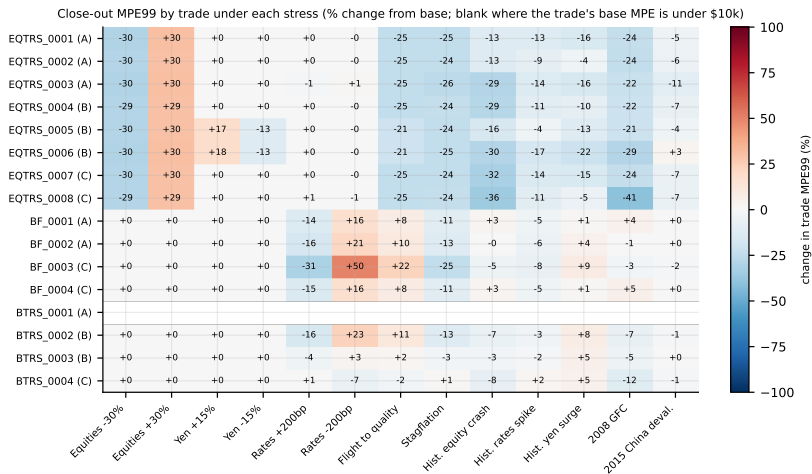


Extended public price history back to 2007 (28 of 37 names covered), found the worst 10-day window inside two named crises:

- **2008 GFC:** Sep 29 – Oct 10, 2008; –23% median / –44% worst single-name return
- **2015 China deval.:** Aug 2015, –6.8% median

Result: both land *inside* the range the existing scenarios already covered – a validation finding, not just more coverage: the pre-existing suite was not blind to the worst episode on record.

Per-trade sensitivity: does every product react as expected



Four independent validation checks

Check	Method	Result
Martingale test	Discounted asset paths should average to today's market price	Pass – no drift or discretisation bug
Parametric VaR benchmark	Delta-normal VaR (bump-and-reprice deltas $\times$ real covariance) vs. full MC P&L, 99%	Parametric \$11.4M vs. MC \$11.7M – ratio 0.98 (expected range 0.3–1.3)
Kupiec backtest, equity (99%, 242 ten-day windows)	Historical exceedances vs. 5,000-scenario prediction	A: 4/2.4, B: 6/2.4, C: 2/2.4 expected – all pass ($p > 0.05$)
Kupiec backtest, rates (99%, 161 windows, 3 tenors)	Same, for the rate factor under both vol sources	0–4 exceedances vs. 1.6 expected at every tenor – all pass
Excel parametric checks	Independent spreadsheet formulas on public data	0.1% (vol calibration), 1–2% (simulated vs. analytic vol), 5–10% (rate vol fit), 7% (equity vega)

Model risk: every input perturbed, one at a time

Perturbation (base portfolio MPE99 = \$51.5M, N=2,000)	Change
Equity/FX vol $\times 2$	+71.3%
Equity correlations, 30% toward 1	+21.8%
Equity/FX vol $\times 1.25$	+19.3%
Rate–equity dependence, both fall together	+18.2%
Rate volatility $\times 1.25$	+9.4%
Rate–equity dependence, offsetting	–15.3%
Mean reversion $a \times 3$	–4.8%
Two-factor rate model (G2++) instead of one-factor	–3.0%

Conclusion: equity volatility, correlation, and the

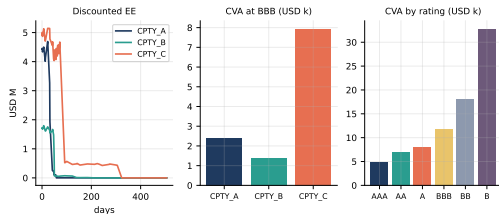
data-unresolvable rate–equity dependence dominate model risk. The one- vs. two-factor rate-model choice explored at length is, by comparison, one of the *smallest* levers – the strongest argument for keeping the simpler model as the default.

Attribution: every correction, one at a time

Step	Portfolio MPE99
Earlier version (flat curve beyond 6.5y, SOFR-overnight σ , no JPY factor, USDJPY drift sign error)	\$45.7M
+ Corrected USDJPY drift sign	\$47.8M
+ Long-end volatility fit and 10y-yield correlations	\$52.7M
+ Bloomberg long end spliced onto the USD curve	\$52.1M
+ JPY Hull-White factor (final model)	\$51.5M

No single correction dominates; each is individually disclosed and traceable – the model-risk study above, not any one fix, is the right lens for “how much could the headline number still move.”

CVA: methodology and results



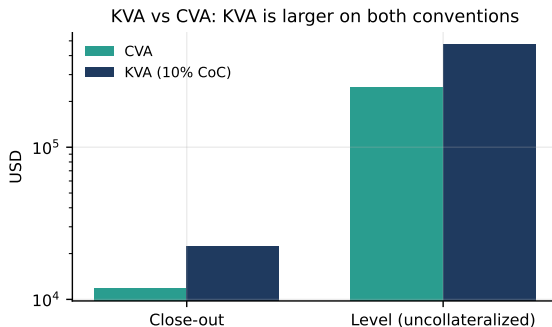
- No CDS available → ICE BofA bond-index OAS proxy by rating (Basel MAR50.32(3)-compliant); all 3 counterparties proxied at BBB
- Credit-triangle hazard-rate survival model:
 $\text{hazard} \approx \text{spread} / (1 - \text{recovery})$, LGD 60%
- CPTY_C close-out CVA \$7.9k; A \$2.4k; B \$1.4k; **total \$11.7k**
- Rating sensitivity: AAA \$5.0k, AA \$7.0k, A \$8.0k, BBB \$11.7k, BB \$18.1k, B \$32.8k

Value delivered: the sensitivity chain (CVA01, the rating table, SA-CVA weighted sensitivities) matters more than the point-in-time number, which is small by construction here.

DVA, FVA, and KVA (extra credit)

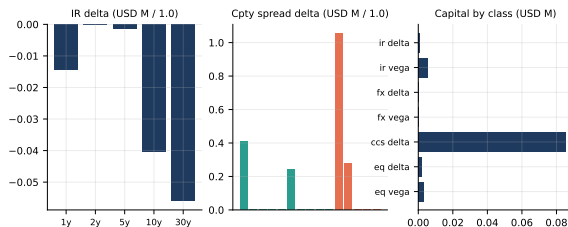
	Close-out	Level	
CVA	\$11.7k	\$247k	On
DVA	\$11.2k	\$7.5k	
FVA	\$0.5k	\$240.0k	
KVA (10% CoC)	\$22.4k	\$470k	

close-out, positive/negative exposure are nearly symmetric (both 10-day moves from a margined start) so CVA/DVA nearly cancel; on level, the book is net in Capitolis' favour, so CVA/FCA dominate.



KVA: $K(t) = 8\% \times RW \times \alpha \times EPE(t)$, cost-of-capital swept 8/10/12% (none given). Exceeds CVA on both conventions because it scales the whole EPE profile by a fixed ratio, not a small IG default probability – explicitly illustrative.

Regulatory capital: SA-CVA and SA-CCR



- SA-CVA capital: \$0.10M close-out / \$2.23M level (RWA \$1.2M / \$27.8M)
- Dominated by counterparty credit-spread risk on both conventions (5% risk weight on a proxy spread)
- SA-CCR exposure at default: \$260M, dominated by the replacement cost of BF_0003

The kickoff brief flagged *“Risk bonds + CDS data for new sample trades”*. Delivered:

- A sample risky-bond forward (25M notional, 5% BBB corporate; not part of the ESF book)
- Priced and simulated with a real issuer credit curve, re-anchored at every simulated node
- Same hazard-rate survival model and bond-index proxy construction as the counterparty CVA

Limitation, disclosed: no real CDS quotes were obtainable for a hypothetical issuer, so the issuer curve is a bond-index proxy, not actual CDS – delivered in spirit, not the full ask.

Feedback tracker: both rounds, all 11 items

Earlier meeting

1. Robust stress testing **Selected**
2. Sensitivities: method, cost, faster alt. **Selected**
3. PCA explainability and speed **Selected**
4. Final results explanation **Selected**
5. CVA walkthrough and set-up **Selected**

This meeting

6. 2008 and more testing periods **Selected**
7. LHS to calculate sensitivities **Selected**
8. KVA addition **Selected**
9. CVA is for sensitivities **Selected**
10. DV01 via a Jacobian **Selected**
11. Excel parametric vol-bump test **Selected**

Detail and results for every item are in the sections above (Data, Models, Methodology, Sensitivities, Stress, Validation, Credit & Capital).

Recap: every trade-off, one line each

Decision	Why (benefit of the choice, cost of the alternative)
One-factor Hull-White, not two-factor	Two-factor moves MPE99 by only -3.0% ; 16.1% fit error; a parameter at its bound; book is a level bet
Dedicated JPY Hull-White factor	Real, independent JPY rate risk incl. negative rates; moved MPE99 \$52.1M $\rightarrow$ \$51.5M; all 3 real mean-reversion routes invalid, so $a=0.001$ used
Full correlation matrix, not PCA	PCA is slower (2.08s vs. 1.91s) at $n=39$ with up to 1.7% extra error; kept as explainability tool only
Latin Hypercube sampling	50.6x lower RMSE than pseudo-random on the controlled test; no consistent Greeks-noise edge on PFE99
Realised long-end vol, not overnight SOFR vol	Matches the actual 10-day long-yield moves this book depends on (15bp vs. 11bp implied); both pass Kupiec
Close-out exposure primary, level secondary	The brief's definition; level is an honest no-margin upper bound (4x larger)
Bump-and-reprice with CRN, pathwise cross-check	282x noise reduction; pathwise 1000x faster for linear risk, validated to 0.02–0.4%
Par-instrument DV01 buckets	Corrected real mis-attribution: 82% vs. 54% of risk correctly placed in the 30y bucket
Exact LHS-reuse rate Greek	Eliminates resimulation cost for curve-only bumps, 10^{-15} validated
Rating-proxy CVA/DVA/FVA/KVA	No CDS/real ratings available; Basel-compliant proxy, fully disclosed, swept over a CoC grid for KVA

Keep as production default

- One-factor Hull-White rate model
- Full correlation matrix (not PCA)
- Latin Hypercube sampling, $N=5,000$ standard / 10,000 sign-off
- Par-instrument DV01 buckets as the primary rate-risk breakdown
- CVA/DVA/FVA/KVA as sensitivity-generating tools first, point estimates second

Open items for the next phase

- Real counterparty ratings or CDS
- Capitolis' own credit and funding spread
- A real capital methodology and cost of capital for KVA
- A decision on wrong-way risk
- The true CSA terms (threshold, MTA, IM)
- Sector-specific (not just rating-based) credit-spread proxy

Thank you

Questions and discussion